Finalization Phase - MongoDB Batch Storage Pipeline

Katleho Mokhele

92124914

DLBDSEDE02

Final Phase

In this final phase, I managed to review and finalize the MongoDB based data storage system, designed to load batches of environmental sensor data for our municipal use case. This data system is built with Docker to enable portability and deployed MongoDB as the database backend due to its document-based schema, flexibility and the ability to scale horizontally.

A sample environmental dataset from Kaggle was acquired, converting it into a JSON format, loading and storing it using a python script that interfaces with MongoDB through PyMongo library. To manage the MongoDB container, I used docker compose to ensure a clean and replicable infrastructure. The architecture is cloud ready and will support future migrations to other environments and due to its flexible schema it can scale as additional sensor data is integrated.

From its conception to its implementation, the solution proved to be a scalable and maintainable data ingestion pipeline. The final implementation is available on GitHub with all the code and setup instructions necessary to replicate the system.

Personal Reflection

This project exposed me to practical data engineering concepts such as NoSQL data modelling, data portability and containerization. Challenges included formatting of raw datasets into a proper JSON structure and management of service connections in docker. I resolved these challenges by leveraging community documentation and testing iterations. This project improved my problem-solving skills and cemented my understanding of portable cloud ready architecture. Going forward , I plan to integrate data validation steps and explore automated ETL framework for large pipelines.

GitHub Repo Link: https://github.com/katleho-mokhele-iu/sensor-data-pipelien.git